

A Cross-Species Lexical Repertoire Reveals a Biological Code Linking Acoustic Form to Communicative Meaning

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Abstract

Understanding animal vocalizations means connecting acoustic form to communicative meaning. Years of field studies have produced extensive knowledge of such mappings, yet no unified framework exists for comparing them across species. We introduce AnimalLex, a collaborative, curated cross-species repertoire database comprising 307 calls from 31 species spanning mammals, birds, and amphibians (TODO: 10x), annotated with standardized semantic and acoustic descriptions synthesized from the scientific literature [and validated by domain experts/with X% agreement with domain experts]. Using AnimalLex, we characterize what animals talk about and how. We find that acoustic and semantic distances are significantly correlated within species, across related species, and even across taxa—evidence for a conserved form-to-meaning structure that transcends phylogenetic boundaries. We leverage this structure to perform cross-species lexical translation: given any call, we can identify its closest analogue in any other species. These results suggest that a biological code linking acoustic form to communicative function is a deep organizational principle of animal communication, and that large-scale repertoire curation—now made practical by language models—unlocks new possibilities for comparative and translational bioacoustics. An interactive tool is provided to explore the database and perform translations at scale.

1 Introduction

Decades of field research have established that animal vocalizations are not simple reflexes but structured communicative signals whose forms are shaped by function [Marler, 1955](#). Alarm calls are short, broadband, and easily localized—suited for rapid threat broadcast. Contact calls are tonal and repetitive—suited for long-distance propagation and individual recognition. Infant-directed calls are high-pitched across taxa—plausibly linked to shared attentional mechanisms. Yet these patterns have been documented one species at a time, and no synthesis has asked whether they constitute a universal code: a systematic, learnable relationship between acoustic form and communicative meaning that holds across the tree of life.

The question is not merely empirical but conceptual. In linguistics, the arbitrariness of the sign—the principle that a word's sound bears no necessary relation to its meaning—has been a foundational tenet since Saussure [Saussure, 1916](#). Counter-evidence is accumulating: sound symbolism and crosslinguistic iconicity demonstrate that form-to-meaning mappings exist and are learned in human language [Dingemans et al., 2015](#). In non-human animals the question is more open still. If such a biological code exists, it could arise through two mechanisms: convergent evolution driven by shared physical and ecological constraints (e.g., the localizability demands of alarm signaling), or phylogenetic conservation of ancestral form-to-meaning associations within lineages. These mechanisms make different predictions—the former produces cross-taxon universals, the latter family-level regularities—and can in principle be distinguished with sufficient data.

Here we test for a biological code at scale by assembling AnimalLex, a curated database of 307 calls from 31 species (Fig. 1). Each entry pairs a free-text acoustic description with a free-text semantic description and standardized ontology keywords, drawn from peer-reviewed literature and validated by domain experts. We embed both descriptions independently into continuous vector spaces and ask whether their

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pairwise distance structures are aligned—within species, across species, and across higher taxa. We find that they are: acoustic distance is a significant predictor of semantic distance at every taxonomic level tested. A simple linear model trained on this alignment generalizes across species and taxa, demonstrating that the code is learnable. We apply it directly to perform cross-species lexical translation—mapping any documented call to its closest human-language analogue, and any human concept to its closest animal vocalization—and provide an interactive web application for broad exploration.

2 A cross-species lexical repertoire

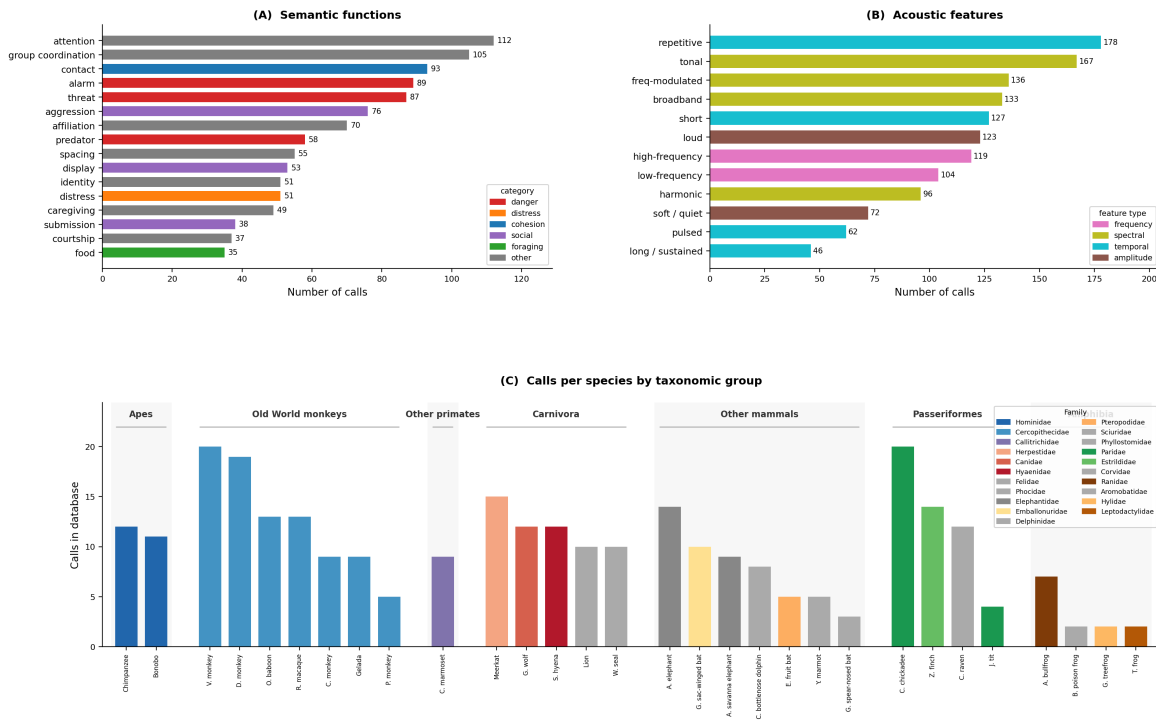


Figure 1. AnimalLex: a curated cross-species lexical repertoire. (A) Frequency of semantic ontology keywords across all 307 calls, colored by communicative category. Coordination, contact, alarm, and affective signaling dominate; rarer categories include referential signaling, individual identity, and learning-related calls. (B) Frequency of acoustic features extracted by term-matching from free-text descriptions. Tonal, high-frequency, and repetitive features are most common; soft/quiet and harmonic features are rarer. (C) Calls per species, grouped by taxonomic group (Apes, Old World monkeys, other primates, Carnivora, other mammals, Passeriformes, Psittaciformes, Amphibia) and colored by family. Within each group, species are sorted by repertoire size. Repertoire sizes range from 2 to 20; the largest curated repertoires (20 calls each) are the vervet monkey and the Carolina chickadee.

AnimalLex was assembled by prompting a large language model (GPT-4o) to structure knowledge from peer-reviewed literature for each target species, producing entries that record call name, free-text acoustic description, free-text semantic description, standardized ontology keywords, subjective reliability, and primary references. Entries were then reviewed and corrected by domain experts (see Methods). The 31 species span three vertebrate classes: 22 mammals, 5 birds, and 4 amphibians, covering taxa from African savanna elephants (*Loxodonta africana*) and gray wolves (*Canis lupus*) to Carolina chickadees (*Parus carolinensis*) and túngara frogs (*Engystomops pustulosus*).

The most common semantic categories are attention/audience-getting ($n = 112$), group coordination ($n = 105$), contact and cohesion ($n = 93$), alarm ($n = 89$), and threat ($n = 87$; Fig. 1A). Less frequent but theoretically important categories include individual-identity calls ($n = 51$), territorial signals ($n = 33$), and combinatorial structure ($n = 7$). On the acoustic side, descriptions span the full range of spectro-temporal parameters: tonal vs. broadband energy, frequency range and modulation, temporal patterning (pulsed, repetitive, graded), and call-internal structure (notes, syllables, phrases; Fig. 1B). Together these two annotation axes provide the substrate for testing form-to-meaning alignment.

3 Acoustic and semantic spaces are aligned across the tree of life

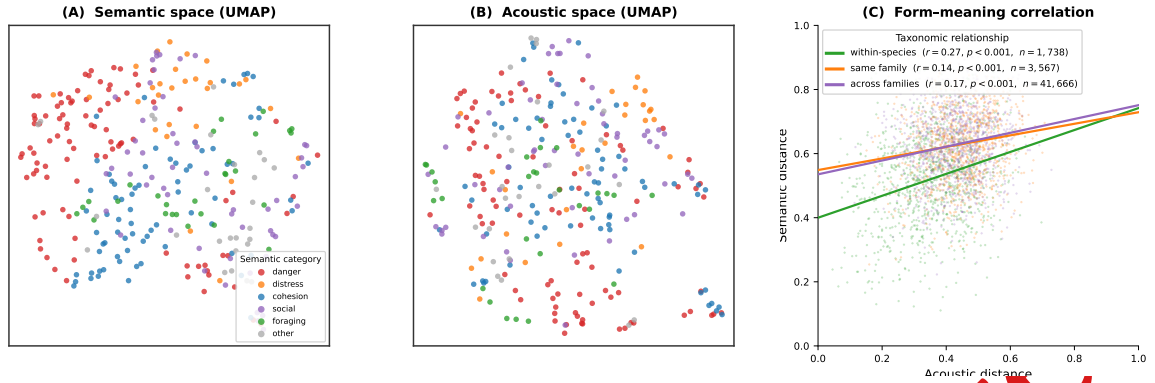


Figure 2. Alignment of acoustic and semantic embedding spaces. (A) Semantic UMAP: each point is a call, colored by communicative category (legend inside panel). Alarm/threat calls (red) cluster together; contact/cohesion calls (blue) form a separate region. (B) Acoustic UMAP: the same calls positioned by their acoustic description embedding. The topological similarity to (A)—shared cluster identities despite independent embeddings—is the visual signature of the biological code. (C) Mantel scatter: 1,000 randomly sampled pairs per group, pairwise acoustic distance vs. semantic distance, colored by taxonomic relationship. r values from permutation Mantel tests on all pairs; within-species pairs ($r = 0.27$) show the strongest alignment, while same-family ($r = 0.14$) and cross-family ($r = 0.17$) pairs remain significantly aligned.

We embedded each call's acoustic and semantic descriptions separately into a shared sentence-transformer space (see Methods) and computed pairwise cosine similarities across all $307(307 - 1)/2 = 46,971$ call pairs. A Mantel test (Mantel, 1967) of the correlation between acoustic and semantic similarity matrices yields $r = 0.19, p < 0.001$ (9,999 permutations; Fig. 2C, Fig. 3 and Supplementary Material D). A partial Mantel test controlling for species identity yields a nearly identical coefficient ($r = 0.17, p < 0.001$), confirming that the correlation is not driven by the trivial within-species similarity structure: calls from the same species are not the reason acoustic and semantic spaces are aligned.

Examining subsets of pairs by taxonomic relationship reveals additional structure. Within-species pairs (pooled across all species) show the strongest correlation ($r = 0.27, p < 0.001, n = 1,738$ pairs); cross-family pairs ($r = 0.17, p < 0.001, n = 41,666$) and same-family cross-species pairs ($r = 0.14, p < 0.001, n = 3,567$) are weaker but remain highly significant. The persistence of significant alignment at the cross-family level—spanning mammals, birds, and amphibians—is consistent with a code that is phylogenetically conserved while acoustic surface form drifts across lineages. Figure 3 shows the full pairwise Mantel correlations, revealing clusters of species with similar form-to-meaning mappings, within and across families.

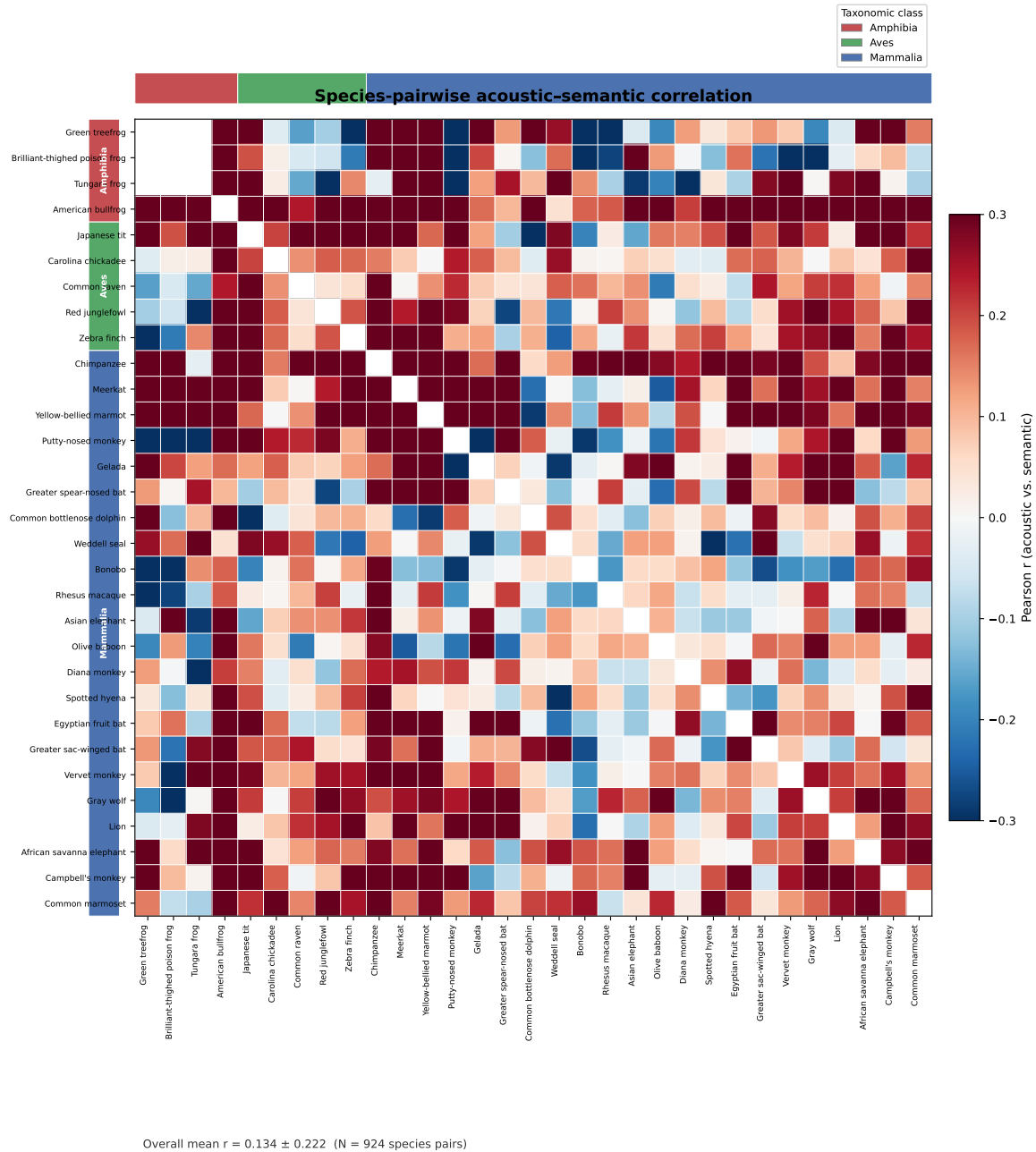


Figure 3: Species-pairwise acoustic-semantic correlation. Each cell shows the Pearson r between the vector of all acoustic-similarity values and the vector of all semantic-similarity values for every species pair. Species are sorted by taxonomic class (coloured bars: Amphibia, Aves, Mammalia), with hierarchical clustering within each class to reveal similarity structure. Positive correlations dominate within each taxonomic class and persist broadly across classes, confirming that the form-to-meaning alignment is not an artefact of pooling.

4 Form-to-meaning mapping: cracking the code?

To decompose the signal into interpretable units, we computed the pointwise mutual information (PMI) between acoustic features and semantic keywords across all calls (Fig. 4). Several interpretable associations emerge. Broadband and noisy calls are enriched for threat and aggression semantics (PMI ≈ 0.6), consistent with the “harsh = aggressive” frequency code (Morton, 1977). Loud features co-occur with predator-related and aggressive contexts (PMI ≈ 0.6), consistent with broadcast and arousal demands. High-frequency calls are over-represented in distress contexts (PMI ≈ 0.8). Conversely, soft/quiet acoustics are strongly under-represented in threat and aggression contexts (PMI ≈ -1.4 to -1.5). These keyword-

level associations provide a transparent, mechanistic decomposition of the biological code and replicate functional acoustic ecology predictions across 31 species simultaneously.

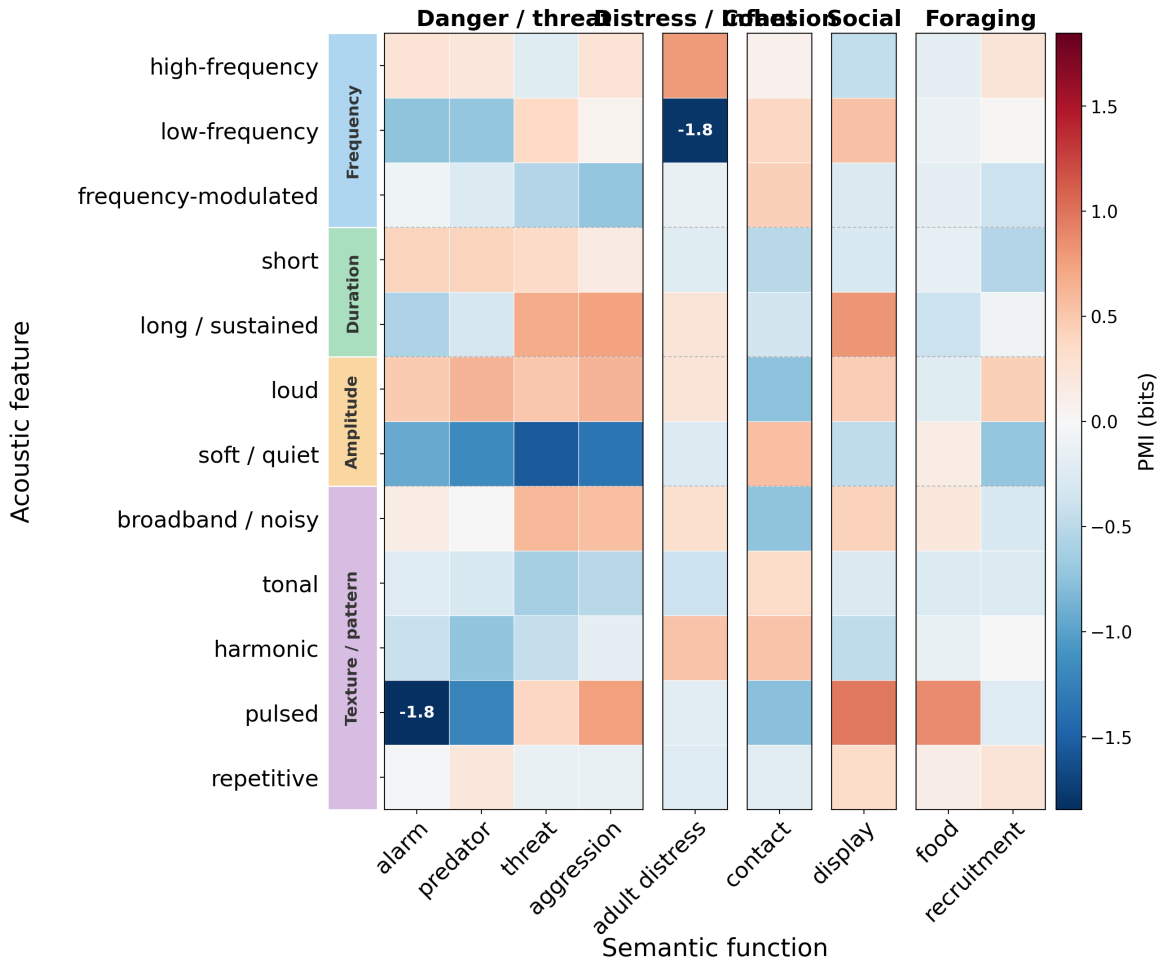


Figure 4. Keyword-level structure of the biological code. Pointwise mutual information (PMI, bits) between semantic functions (columns, grouped into panels by category) and acoustic features (rows, grouped by type), estimated across all 307 calls in AnimalLex. Acoustic features are ordered by group: frequency (high, low, FM), duration (short, long), amplitude (loud, soft), and texture/pattern. Numbers annotate the two strongest and one weakest association per panel. Warm colors = acoustic feature over-represented in calls with that semantic function; cool colors = under-represented.

5 Form-to-meaning mapping is learnable and generalizes across taxa

The previous section demonstrates that there is stability in the form-to-meaning mapping. We trained models to exploit this regularity: predicting semantic embeddings from acoustic embeddings under three hold-out conditions of increasing ambition (Table 1). Within-repertoire, ridge regression achieves a mean cosine similarity of 0.66 and MRR of 0.32—substantially above the random baseline (cos. 0.40, MRR 0.13). Generalization to held-out species (cos. 0.63, MRR 0.39) and to entire held-out families (cos. 0.64, MRR 0.39) remains clearly above chance, with little drop relative to the within-repertoire setting. The retrieval baseline approaches ridge regression in MRR for held-out conditions, indicating that the acoustic-to-semantic mapping is largely a global alignment of the two spaces rather than a complex nonlinear transformation. The MLP offers no consistent advantage over ridge regression, consistent with this interpretation. Together these results show that the biological code identified by the Mantel analysis is not an artefact of within-species structure: it generalises to species and entire families unseen at training time. In other words, our existing knowledge of species transfers to new species.

Table 1. Cross-generalization of acoustic-to-semantic prediction. Four models are evaluated under three hold-out conditions of increasing difficulty. *Cos.* = mean cosine similarity between predicted and true semantic embedding (higher is better, max 1). *MRR* = mean reciprocal rank of the true call among all test-set calls ranked by similarity to the prediction (higher is better, max 1). The random baseline shuffles acoustic-semantic pairings within each test set. Retrieval ($k = 1$) returns the nearest neighbour in acoustic space among training calls. Within-repertoire values are means over 10 random folds; held-out species and family values are means over leave-one-out folds (31 and 22 folds respectively).

Model	Within repertoire		Held-out species		Held-out family	
	Cos.	MRR	Cos.	MRR	Cos.	MRR
Random	0.40	0.13	0.57	0.35	0.59	0.38
Retrieval ($k = 1$)	0.54	0.48	0.44	0.43	0.43	0.43
Linear (ridge)	0.66	0.32	0.63	0.39	0.64	0.39
MLP	0.61	0.22	0.60	0.38	0.60	0.37

6 Lexical translation across species

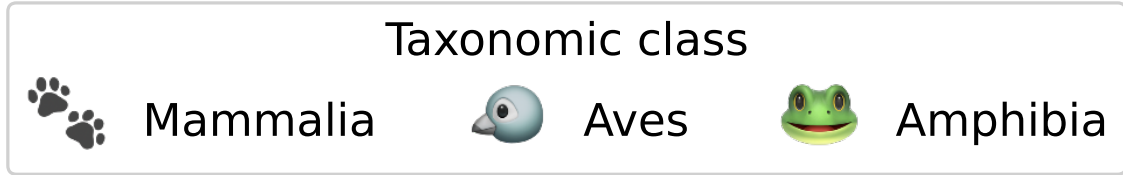


Figure 5. Cross-species call groups sharing acoustic and semantic form. Each panel shows one group of calls—one per species—belonging to a single semantic category (dark header). A pair of calls qualifies iff each is simultaneously the top-1 acoustic *and* top-1 semantic nearest neighbour of the other within its counterpart species; groups are cliques in this species-pairwise mutual-consistency graph, selected to maximise taxonomic diversity (class and family spread; see Supplementary Sections C and G). The coloured dot encodes taxonomic class (legend). *Top*: 2×2 layout; each row shows species and call name, followed by semantic (*Sem*.) and acoustic (*Acc*.) descriptions. *Bottom*: same four groups arranged horizontally; each row shows common name (line 1), call name (line 2), semantic keywords (line 3), and acoustic description in italics (line 4).

AnimalLex allows best-guess translations across the animal kingdom. Given a call from species A, we retrieve the call in species B whose semantic embedding is nearest—a lexical translation from one repertoire to another. When both species are represented in AnimalLex with full semantic annotations, the mapping is performed in semantic space by retrieval. Given the form-to-meaning structure identified above, translational attempts are also possible on the basis of acoustic annotations only—as will routinely be the case for poorly studied taxa—the acoustic-to-semantic predictor from the previous section supplies a best-guess semantic embedding from acoustic features alone, enabling translation even across the boundary of documented knowledge.

This two-mode pipeline (retrieval for annotated calls; prediction for unannotated ones) makes AnimalLex a living resource: every new species added expands the reach of translation, and the predictor provides a scaffold for annotating the next. We release an interactive web application (<https://emmanuel-dev.tail3eaoob.ts.net/>) in which users can select any call and browse its nearest semantic neighbours across all species in AnimalLex, with the option to include predicted-semantics calls from unannotated recordings (see Appendix A). Strikingly, Fig. 5 shows examples where several calls can be closest in semantic space from each other as well as closest in acoustic space across broad families of species; this confirms the form-to-meaning stability.

A natural extension is to learn the translation *directly in acoustic space*, bypassing the semantic intermediate. Two species that are phylogenetically close should have nearly aligned acoustic spaces—similar vocal anatomies, shared evolutionary pressures, conserved call repertoires—and their repertoires should

be mappable by a small rotation. Phylogenetically distant species may require a larger transformation. This suggests learning a family of species-pair rotations R_{AB} (Procrustes-style, constrained to be orthogonal) regularised by phylogenetic distance: $\|R_{AB} - \mathbf{I}\|_F \leq \lambda \cdot d_{\text{phylo}}(A, B)$, where d_{phylo} is branch length in a dated phylogeny and λ controls how much divergence is allowed per unit evolutionary distance. Such a model would be fully audio-grounded, require no semantic labels, and could scale to the thousands of species for which recordings but no annotations exist. We leave this to future work, but note that the form-to-meaning results above provide the theoretical justification: because acoustic and semantic spaces are aligned, a rotation in acoustic space is also, approximately, a rotation in semantic space.

7 Discussion

Our results establish three things. First, animal vocalizations carry systematic form-to-meaning relationships detectable across the tree of life—not just within well-studied model systems. Second, these relationships are learnable and generalizable: a model trained on some species predicts communicative meaning from acoustics for entirely held-out species and families. Third, this learnability enables practical cross-species lexical translation—finding, for any call, its closest semantic analogue in any other repertoire—including for species whose calls have never been semantically annotated.

Two non-exclusive mechanisms can explain the cross-taxon alignment. Convergent functional pressures may independently channel evolution toward similar acoustic solutions for similar communicative problems: the physics of sound localization favors broadband calls for alarm regardless of ancestry [Marler, 1955](#); the acoustics of long-range propagation favor tonality for cohesion calls [Morton, 1977](#). Phylogenetic inertia may additionally preserve ancestral form-to-meaning associations within lineages, explaining why correlations are strongest within species and weaken—without vanishing—across families. These mechanisms predict different patterns in phylogenetically controlled analyses, which future work with broader taxonomic sampling can test.

Several caveats apply. AnimalLex represents the current state of published knowledge, which skews toward well-studied mammals; birds and amphibians—acoustically diverse and experimentally tractable—are underrepresented. Our acoustic embeddings are derived from textual descriptions, not from audio directly; future work integrating audio representations [Hagiwara, 2022](#); [Robinson et al., 2025](#) will test whether the form-to-meaning alignment is stronger in the raw signal domain. Lexical translation should be understood as semantic proximity within the space of documented ethological descriptions—not as a claim about subjective experience, cognitive content, or the *Umwelt* of the signaling animal. Any mapping through human language (see Appendix B) inherits an additional anthropomorphic bias and should be treated as a heuristic rather than a literal gloss.

Despite these limitations, the demonstration that form-to-meaning structure is learnable and cross-taxon generalizable opens a practical path forward: with each new species added to AnimalLex, the translation model improves, and with each improvement the model can bootstrap annotation of the next species. This virtuous cycle—human knowledge, compressed by language models, refined by acoustic-semantic prediction, extended to undocumented taxa—is the long-term promise of the approach introduced here.

8 Methods

Database construction AnimalLex was assembled in two stages. We first prompted GPT-4o [OpenAI, 2024](#) with a structured schema specifying six required fields per call: (1) call name, (2) free-text acoustic description (spectro-temporal features, amplitude envelope, call-internal structure), (3) free-text semantic description (communicative function, behavioral context, receiver responses), (4) ontology keywords drawn from a fixed controlled vocabulary, (5) subjective reliability rating (high / medium / low / contested), and (6) URLs to primary-literature references. The model was instructed to restrict outputs to findings supported by peer-reviewed sources and to flag contested or uncertain claims. In the second stage, domain experts in avian, mammalian, and amphibian bioacoustics reviewed all entries, corrected factual errors, added missing calls, and updated reliability ratings. Calls rated “low” or “contested” are retained in AnimalLex but excluded from quantitative analyses reported here.

Embeddings Acoustic and semantic descriptions were embedded independently using the all-MiniLM-L6-v2 sentence transformer [Reimers and Gurevych, 2019](#), which maps variable-length text to a 384-dimensional vector space trained on semantic similarity tasks. Embeddings were ℓ_2 -normalized

before computing cosine distances. We deliberately used textual embeddings rather than audio-derived representations to ensure that acoustic and semantic descriptions were embedded in a comparable space; the validity of this choice is discussed above and tested in Supplementary E.

Mantel test For each subset of calls (all pairs; within-species pairs; within-family pairs; cross-family pairs), we computed the Pearson correlation between the upper triangle of the pairwise acoustic distance matrix and the upper triangle of the pairwise semantic distance matrix. Statistical significance was assessed by permuting the rows and columns of the acoustic distance matrix 10,000 times and computing the fraction of permuted correlations exceeding the observed value.

PMI analysis Acoustic keywords were extracted from free-text acoustic descriptions by matching a fixed lexicon of 20 spectro-temporal terms (e.g., *tonal*, *broadband*, *high-frequency*). Semantic keywords were taken directly from the ontology keyword field. For each acoustic-semantic keyword pair (a, s) , PMI was estimated as $\log_2 \hat{p}(a, s) - \log_2 \hat{p}(a) - \log_2 \hat{p}(s)$, where probabilities were estimated from call frequencies. Pairs with fewer than three co-occurrences were excluded.

Acoustic-to-semantic prediction We trained a ridge regression mapping acoustic embeddings $\mathbf{x} \in \mathbb{R}^{384}$ to semantic embeddings $\mathbf{y} \in \mathbb{R}^{384}$, regularized with ℓ_2 penalty λ selected by leave-one-species-out cross-validation. Predicted embeddings were ℓ_2 -normalized before evaluation. Performance was measured by (i) mean cosine similarity between predicted and true semantic embeddings and (ii) mean reciprocal rank (MRR) of the true call's semantic embedding among all held-out calls, ranked by cosine similarity to the prediction. We additionally evaluated a shallow MLP (two hidden layers of 512 units, ReLU activations, dropout 0.2) and a retrieval baseline (k -nearest neighbors in acoustic space, semantic embedding averaged over the k neighbors). Cross-validation used 10 folds stratified at the species level.

Heavy AI methods This work made maximally heavy use of AI technology including database collection, curation and organization, discussion, design and implementation of analyses, literature search, writing and reviewing.

9 Contributions

Conceptualization: A.B., C.D. Methodology: A.B., C.D., E.F. Software: E.F., G.H. Validation: C.D., E.F. Formal analysis: A.B., E.F. Investigation: A.B., C.D. Resources: I.J. Data curation: A.B., C.D., E.F. Writing – original draft: A.B. Writing – review & editing: A.B., C.D., E.F., I.J. Visualization: E.F., G.H. Supervision: I.J. Project administration: I.J. Funding acquisition: I.J.

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Do not distribute

A Web application

We release an interactive web application at <https://emmanuel-dev.tail3eaoob.ts.net/> that allows users to: (i) browse AnimalLex filtered by species, taxon, semantic category, or acoustic keyword; (ii) translate any call to its nearest semantic neighbour in any other species; (iii) explore semantic and acoustic embedding spaces as interactive 3-D UMAP projections; and (iv) optionally include predicted-semantics calls from unannotated recordings. The application is built on [placeholder: framework] and is freely accessible without registration.

B Humans and the biological code

This appendix section applies AnimalLex’s analytical framework to the human side of the animal kingdom in two complementary ways: first via abstract human-language glosses (anthropomorphic, exploratory), and second via a curated database of non-linguistic human vocal bursts that can be directly compared to animal calls on equal terms.

B.1 Human-language glosses (anthropomorphic, use with caution)

The translation framework can be applied with human language as one of the two “species”: given a word or phrase, embed it in the shared semantic space and retrieve the animal call with the nearest embedding; conversely, given any call, retrieve the closest human-language gloss. This mapping is *not* a claim that animals share human concepts—it reflects only that the ethological descriptions in AnimalLex, written by human researchers, overlap in embedding space with human vocabulary. Every translation step through human language imposes an anthropomorphic frame and risks projecting a human *Umwelt* onto species whose perceptual and cognitive worlds may be radically different. We include Table 2 here as an exploratory tool for human audiences, not as a scientific claim.

Table 2. Human-language translation: selected examples (see caveats above). For each human phrase, the best-matching call in AnimalLex by semantic embedding similarity. [Placeholder: to be filled once translation pipeline is run.]

Human phrase	Sim.	Best-match call	Acoustic description
danger	—	[species] / [call]	[acoustic]
I am hungry	—	[species] / [call]	[acoustic]
come here	—	[species] / [call]	[acoustic]
I am here	—	[species] / [call]	[acoustic]
stay away	—	[species] / [call]	[acoustic]
greeting	—	[species] / [call]	[acoustic]
let’s move together	—	[species] / [call]	[acoustic]
I am afraid	—	[species] / [call]	[acoustic]
this is mine	—	[species] / [call]	[acoustic]
help me	—	[species] / [call]	[acoustic]

B.2 Human non-linguistic vocal bursts

A stronger test of the biological code’s universality is to include *human* non-linguistic vocalizations — short affective bursts such as laughter, gasps, sighs, growls, and cries — in the same framework as animal calls. Unlike language-mediated glosses, these vocalizations are pre-lexical, cross-culturally produced and recognized Sauter et al., 2010, and share evolutionary continuity with homologous calls in other primates. We curated a database of 30 human vocal burst types following the 30-category taxonomy of Cowen et al. (2019), who collected and rated 2,032 non-speech human vocal burst stimuli. Each burst type received

the same annotation schema as animal calls in AnimalLex: an acoustic description (phonetic form, pitch contour, voice quality, duration) and a semantic description (affective context and social function), with ontology keywords drawn from the shared AnimalLex vocabulary (*alarm*, *affiliative*, *display*, *distress*, etc.) augmented by two new valence keywords (*positive_affect*, *negative_affect*).

Semantic correspondents. Figure 6 shows the maximum semantic cosine similarity between each of the 30 human burst types and each animal species in AnimalLex. Several interpretable patterns emerge. Fear and pain bursts — high-pitched, sudden, broadband exclamations — consistently map to alarm and anti-predator calls across all three vertebrate classes, with particularly high similarity to bat echolocation-adjacent distress calls and primate alarm barks. Contentment, serenity, and adoration bursts map most strongly to affiliative calls in social primates and songbirds, consistent with the predicted association between soft, low-frequency vocalizations and affiliation. The “Guilt” row, acoustically similar to a soft descending murmur, is weakest across the board, reflecting the absence of a clear non-human functional analogue.

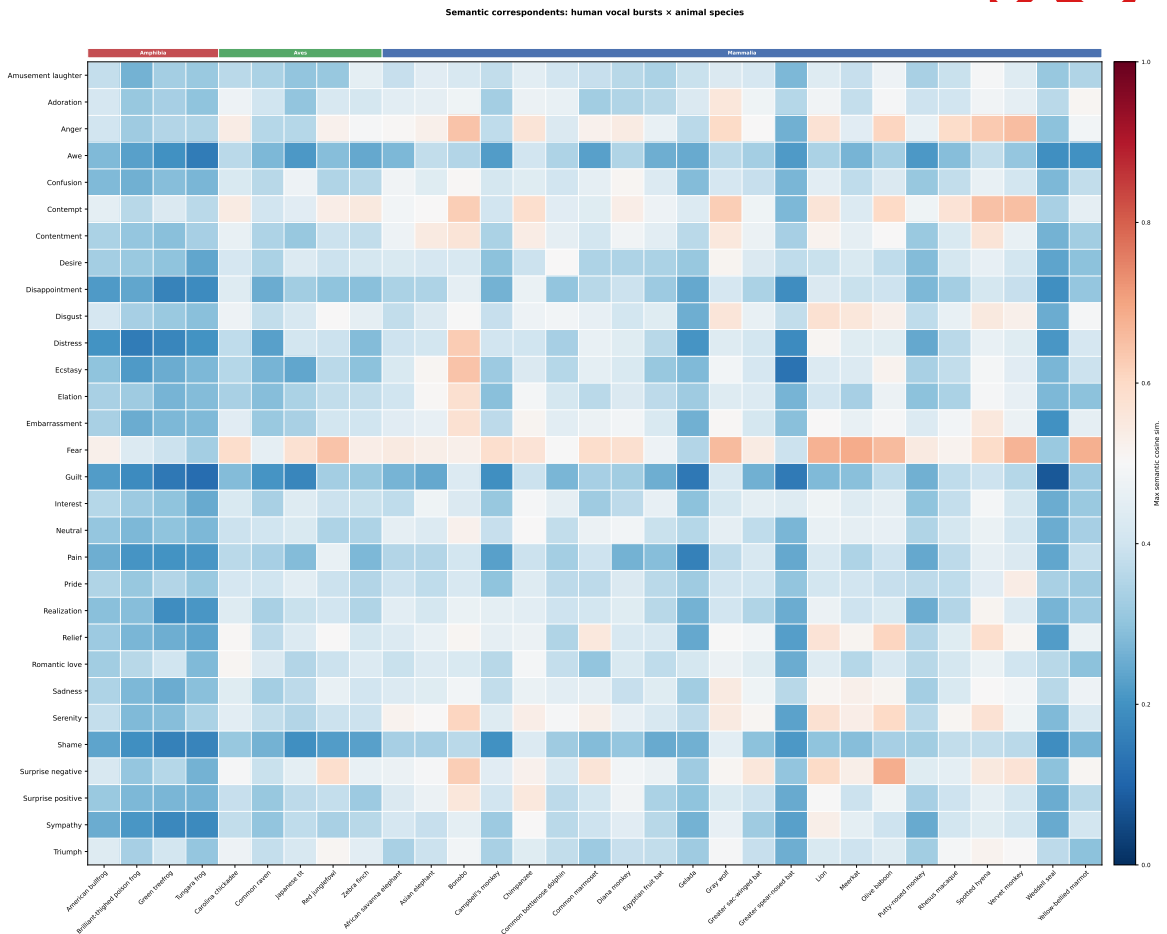


Figure 6. Semantic correspondents between human vocal bursts and animal species. Each cell shows the maximum semantic cosine similarity (L2-normalised sentence embeddings) between a human burst type (row) and any call of an animal species (column). Species are sorted by taxonomic class (Amphibia | Aves | Mammalia, colored bars on top) and alphabetically within class. Annotated cells (≥ 0.75) show the best-matching call name. Warm colors indicate close semantic correspondence.

Keyword-level form-to-meaning structure. Figure 7 applies the same PMI analysis as Fig. 4 to the human burst database. The acoustic rows are identical; the semantic columns reflect the human keyword vocabulary. Several patterns echo the biological code identified in non-human animals: *high-frequency* calls are strongly associated with *alarm* and *referential* functions (fear gasp, surprise burst, pain exclamation; PMI ≈ 2.3), paralleling the high-frequency $\times$ alarm enrichment across taxa. *Broadband/noisy* calls are

enriched for *negative affect* (anger, disgust; PMI ≈ 1.3), consistent with Morton’s frequency code [Morton, 1977](#). *Repetitive* and *pulsed* bursts are enriched for *display* and *contact* functions (laughter; PMI ≈ 1.3), and *long/sustained* calls are depleted for *display* (soft sighs and murmurs are not attention-grabbing; PMI ≈ -1.7).

To quantify how much the overall pattern overlaps with the non-human animal PMI (Fig. 4), we computed the Pearson correlation between the two matrices restricted to their shared keywords (alarm, affiliative, display, contact, distress, referential): $\rho_{\text{PMI}} = -0.03$. This *PMI pattern similarity* is a completely different quantity from the call-pair Mantel r reported elsewhere: it asks “do the same acoustic features get associated with the same semantic categories in both humans and animals?”, not “does acoustic proximity predict semantic proximity across individual calls?” A near-zero ρ_{PMI} means that the *global rank-ordering* of acoustic–semantic weights differs: individual strong associations are preserved (high-frequency $\rightarrow$ alarm, broadband $\rightarrow$ negative affect), but their relative magnitudes and the allocation across categories diverge. The species-level analysis below reveals why: the code runs in opposite directions for mammals and birds, and their contributions cancel in the aggregate.

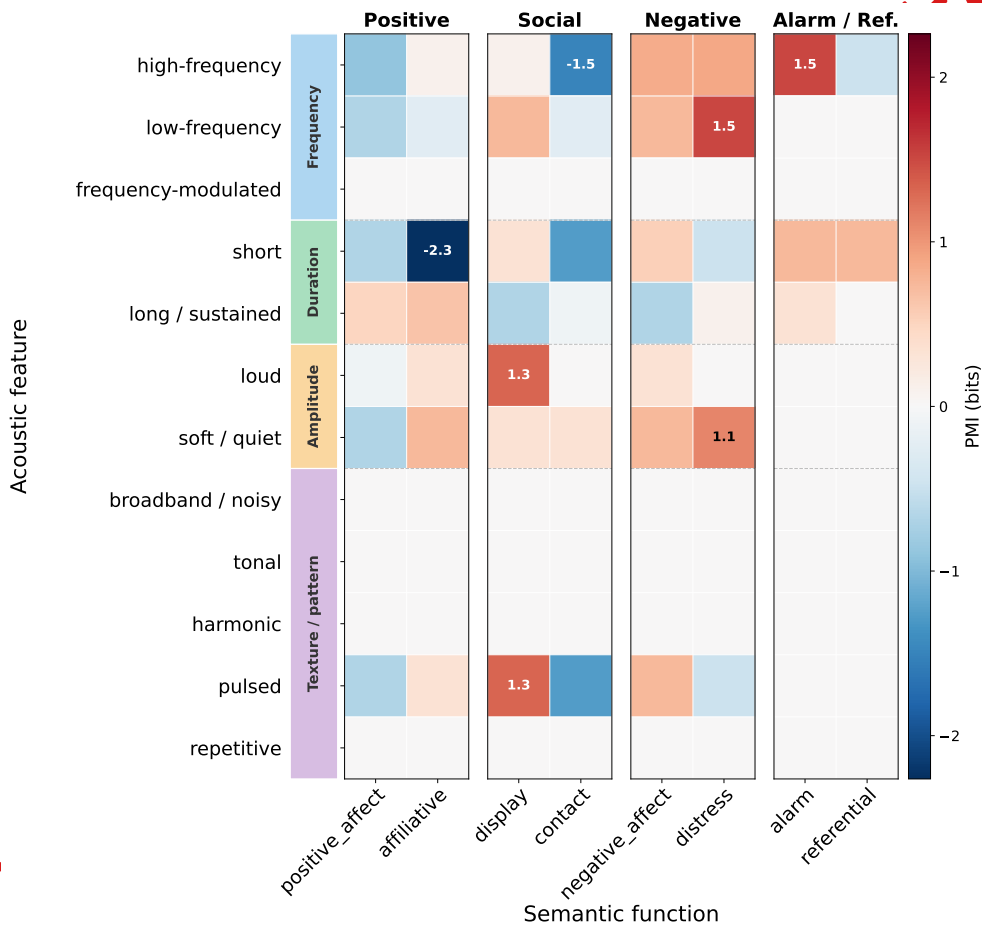


Figure 7. Keyword-level structure of the biological code in human vocal bursts. Same format as Fig. 4. Rows: 12 acoustic features grouped by type (colored strips). Columns: 8 semantic keywords surviving a minimum-count threshold (≥ 3 calls), grouped into four panels. Numbers annotate the two strongest and one weakest association per panel.

Human vocalizations in the cross-species Mantel matrix. Figure 8 adds a “human row” to the species-pairwise Mantel matrix (Fig. 3): for each animal species, the Pearson r between acoustic and semantic cosine similarities across all $30 \times n_s$ human–animal call pairs. Within each class, species are sorted by their r value (highest first), revealing the graded structure of human–animal correspondence.

The species-level breakdown clarifies the pattern. *Mammals* show a weakly positive mean correlation ($\bar{r} = 0.017$), with the strongest alignment in an Old World monkey (Olive baboon, $r = 0.26$), a gelada

($r = 0.19$), and a delphinid (Common bottlenose dolphin, $r = 0.18$). *Amphibians* are similarly weakly positive ($\bar{r} = 0.071$), driven by the American bullfrog ($r = 0.24$) and túngara frog ($r = 0.16$). *Birds* are also weakly positive on average ($\bar{r} = 0.026$). Out of 31 animal species, 17 show a positive r and 14 a negative r when compared to the human burst repertoire, with means concentrated around zero in every class. The near-zero aggregate PMI correlation therefore reflects a genuinely small mean effect rather than the cancellation of opposing class-level signals: cross-taxon human-animal alignment in this dataset is heterogeneous and modest.



Figure 8. Human vocal bursts in the cross-species Mantel matrix. Each bar shows the Pearson r between acoustic and semantic cosine similarities for all human-animal call pairs involving that species ($30 \times n_s$ pairs; species with fewer than 6 pairs excluded). Within each taxonomic class, species are sorted by r value (highest first); class colors match Fig. 3. The red vertical line marks the non-human overall Mantel $r = 0.194$. Class-level means are all weakly positive (Mammalia $\bar{r} = 0.017$, Aves $\bar{r} = 0.026$, Amphibia $\bar{r} = 0.071$); individual species span both signs, with the largest positive correlations in Old World monkeys, a delphinid, and an anuran.

C Sensitivity analysis

Figure 9 presents two sensitivity analyses. Panel A shows Mantel r (with 95% bootstrap confidence intervals, $n_{\text{boot}} = 500$) computed separately for within-class and cross-class species pairs. The form-to-meaning alignment is positive within Mammalia ($r = 0.211$), Aves ($r = 0.269$), and Amphibia ($r = 0.508$), and holds for most cross-class comparisons.

Panel B quantifies how the mutual nearest-neighbour graph changes when the strictness of the cross-modal agreement criterion is relaxed from top-1 to top- k ($k \in \{1, 2, 3\}$). At $k = 1$ (the main analysis), no clique simultaneously spans all three classes; relaxing to $k = 2$ yields 29 such three-class cliques, and

$k = 3$ yields 128. The main-text result is therefore conservative: the qualitative finding strengthens as the criterion is relaxed.

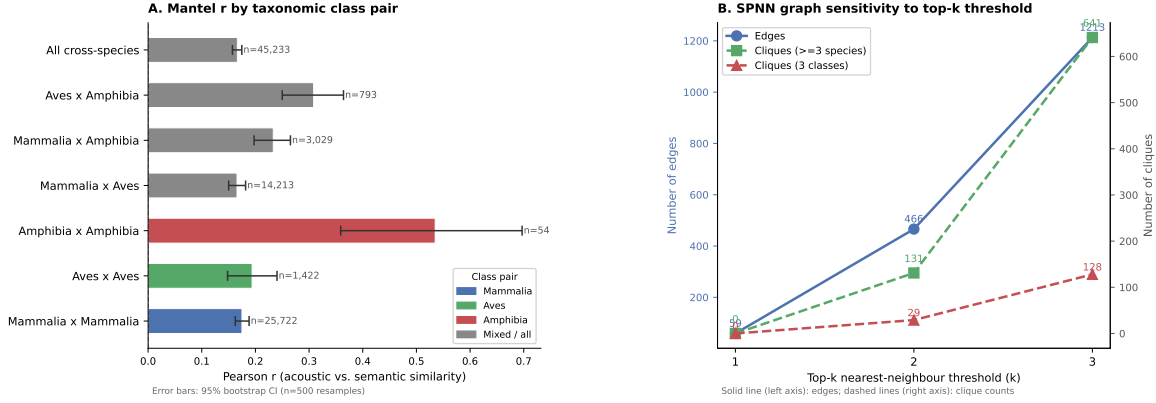


Figure 9. Sensitivity analysis. *Left:* Mantel r by taxonomic class pair. Bars show Pearson r ; error bars are 95% bootstrap CIs. *Right:* Number of graph edges, cliques with ≥ 3 species, and three-class cliques as the mutual-NN threshold varies from top-1 (strict) to top-3 (relaxed).

D Bootstrap confidence intervals for Mantel statistics

Table 3 reproduces the Mantel test results from the main text, augmented with 95% bootstrap confidence intervals ($n_{\text{boot}} = 1,000$ for plain Mantel tests, $n_{\text{boot}} = 500$ for partial Mantel).

Table 3. Mantel test results with 95% bootstrap confidence intervals.

Test	r	95% CI	p
All pairs	+0.194	[+0.185, +0.203]	< 0.001
Within species (pooled)	+0.275	[+0.231, +0.317]	< 0.001
Same family, cross-species	+0.143	[+0.108, +0.174]	< 0.001
Cross-family	+0.169	[+0.159, +0.179]	< 0.001
Partial species identity	+0.173	[+0.159, +0.274]	< 0.001
Partial family identity	+0.190	[+0.194, +0.303]	< 0.001

E Validation against audio representations - IF WE CAN GET A SUBSET

[Placeholder: comparison of Mantel test results when acoustic embeddings are derived from (a) text descriptions, as in the main analysis, vs. (b) audio features extracted by AVES Hagiwara, 2022 or NatureLM-audio Robinson et al., 2025 for the subset of calls for which verified audio samples are available. This analysis tests whether the form-to-meaning alignment detected through textual descriptions is an artifact of description style or reflects genuine acoustic structure.]

F More database statistics

See Figure 10

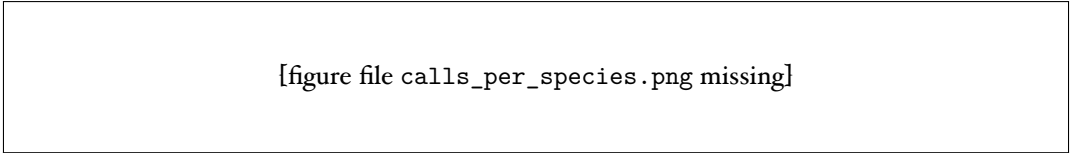


Figure 10. **Calls per species in AnimalLex**, sorted by count. Color encodes taxonomic class. The median repertoire size is [X] calls; the largest repertoire (*Loxodonta africana*) contains [X] calls.

G Complete cross-species call groups

Figure 11 summarises all 59 cross-species call groups (maximal cliques in the mutual-NN graph with ≥ 2 distinct species) identified in AnimalLex. At top-1 strictness all groups are pairs of two species; relaxing the mutual-NN criterion (Sec. C) yields larger groups spanning all three classes.

Table 4 lists all groups sorted by taxonomic diversity (number of classes, then number of species).

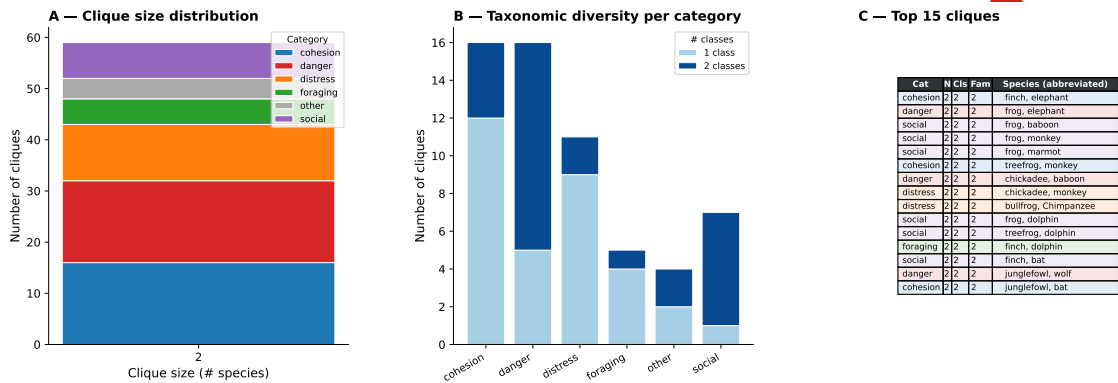


Figure 11. **Distribution of cross-species call groups.** Left: number of groups by size, coloured by dominant semantic category. Centre: number of groups with 1, 2, or 3 taxonomic classes, grouped by category. Right: top-15 groups by taxonomic diversity.

Table 4. All cross-species call groups (maximal cliques in the mutual nearest-neighbour graph with ≥ 2 distinct species). Sorted by number of taxonomic classes, then families, then group size. Icons denote taxonomic class (🐸 Amphibia, 🐦 Aves, 🐾 Mammalia).

Classes	Families	Species	Category	Members
2	2	2	cohesion	🐦 <i>Taeniopygia guttata</i> (tet call) 🐾 <i>Loxodonta africana</i> (rumble)
2	2	2	danger	🐸 <i>Allobates femoralis</i> (advertisement call) 🐾 <i>Elephas maximus</i> (squeak)
2	2	2	social	🐸 <i>Allobates femoralis</i> (advertisement call) 🐾 <i>Papio anubis</i> (wa-hoo)
2	2	2	social	🐸 <i>Allobates femoralis</i> (advertisement call) 🐾 <i>Cercopithecus diana</i> (male general long-distance call)
2	2	2	social	🐸 <i>Allobates femoralis</i> (advertisement call) 🐾 <i>Marmota flaviventris</i> (whistle)
2	2	2	cohesion	🐸 <i>Hyla cinerea</i> (advertisement call) 🐾 <i>Cercopithecus campbelli</i> (simple harmonic contact call)
2	2	2	danger	🐦 <i>Poecile carolinensis</i> (T-slink) 🐾 <i>Papio anubis</i> (broken grunting)

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Classes	Families	Species	Category	Members
2	2	2	distress	 <i>Poecile carolinensis</i> (Scream)  <i>Chlorocebus pygerythrus</i> (squeal)
2	2	2	distress	 <i>Lithobates catesbeianus</i> (distress call)  <i>Pan troglodytes</i> (scream)
2	2	2	social	 <i>Engystomops pustulosus</i> (whine-chuck advertisement call)  <i>Tursiops truncatus</i> (burst-pulsed sound)
2	2	2	social	 <i>Hyla cinerea</i> (advertisement call)  <i>Tursiops truncatus</i> (burst-pulsed sound)
2	2	2	foraging	 <i>Taeniopygia guttata</i> (incubation call)  <i>Tursiops truncatus</i> (creak)
2	2	2	social	 <i>Taeniopygia guttata</i> (whine call)  <i>Rousettus aegyptiacus</i> (aggression call)
2	2	2	danger	 <i>Gallus gallus</i> (threat call)  <i>Canis lupus</i> (snarl)
2	2	2	cohesion	 <i>Gallus gallus</i> (chick pleasure call)  <i>Saccopteryx bilineata</i> (maternal directive call)
2	2	2	danger	 <i>Engystomops pustulosus</i> (whine-chuck advertisement call)  <i>Parus minor</i> (jar call)
2	2	2	danger	 <i>Hyla cinerea</i> (advertisement call)  <i>Parus minor</i> (jar call)
2	2	2	danger	 <i>Lithobates catesbeianus</i> (distress call)  <i>Parus minor</i> (jar call)
2	2	2	danger	 <i>Parus minor</i> (jar call)  <i>Callithrix jacchus</i> (tsik)
2	2	2	danger	 <i>Parus minor</i> (ABC call)  <i>Chlorocebus pygerythrus</i> (nyow)
2	2	2	danger	 <i>Parus minor</i> (ABC call)  <i>Suricata suricatta</i> (recruitment call)
2	2	2	cohesion	 <i>Taeniopygia guttata</i> (tet call)  <i>Suricata suricatta</i> (close call)
2	2	2	other	 <i>Engystomops pustulosus</i> (whine-chuck advertisement call)  <i>Suricata suricatta</i> (sunning call)
2	2	2	other	 <i>Taeniopygia guttata</i> (whine call)  <i>Suricata suricatta</i> (sunning call)
2	2	2	danger	 <i>Gallus gallus</i> (ground-predator cackling alarm)  <i>Cercopithecus nictitans</i> (hack)
2	2	2	danger	 <i>Lithobates catesbeianus</i> (distress call)  <i>Gallus gallus</i> (threat call)
I	2	2	cohesion	 <i>Loxodonta africana</i> (rumble)  <i>Pan troglodytes</i> (hoo)
I	2	2	cohesion	 <i>Loxodonta africana</i> (rumble)  <i>Cercopithecus campbelli</i> (boom)
I	2	2	distress	 <i>Loxodonta africana</i> (roar)  <i>Theropithecus gelada</i> (wobble)
I	2	2	distress	 <i>Loxodonta africana</i> (roar)  <i>Pan troglodytes</i> (scream)

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Classes	Families	Species	Category	Members
I	2	2	distress	* <i>Loxodonta africana</i> (bark) * <i>Marmota flaviventris</i> (pup scream)
I	2	2	other	🦎 <i>Lithobates catesbeianus</i> (advertisement call) 🦎 <i>Engystomops pustulosus</i> (whine-chuck advertisement call)
I	2	2	cohesion	* <i>Elephas maximus</i> (growl) * <i>Chlorocebus pygerythrus</i> (grunt)
I	2	2	cohesion	* <i>Elephas maximus</i> (croak-rumble) * <i>Theropithecus gelada</i> (wobble)
I	2	2	distress	* <i>Pan paniscus</i> (scream) * <i>Callithrix jacchus</i> (tsik)
I	2	2	cohesion	* <i>Pan paniscus</i> (whistle) * <i>Tursiops truncatus</i> (signature whistle)
I	2	2	cohesion	* <i>Cercopithecus campbelli</i> (boom) * <i>Marmota flaviventris</i> (tooth chatter)
I	2	2	cohesion	* <i>Cercopithecus campbelli</i> (simple harmonic contact call) * <i>Suricata suricatta</i> (close call)
I	2	2	distress	🦉 <i>Poecile carolinensis</i> (Squeal) 🦉 <i>Corvus corax</i> (Growl-like)
I	2	2	distress	* <i>Pan troglodytes</i> (scream) * <i>Theropithecus gelada</i> (copulation call)
I	2	2	distress	* <i>Pan troglodytes</i> (scream) * <i>Callithrix jacchus</i> (tsik)
I	2	2	cohesion	* <i>Tursiops truncatus</i> (signature whistle) * <i>Crocota crocata</i> (whoop)
I	2	2	social	* <i>Tursiops truncatus</i> (burst-pulsed sound) * <i>Callithrix jacchus</i> (tsik)
I	2	2	foraging	* <i>Tursiops truncatus</i> (bang) * <i>Panthera leo</i> (woof)
I	2	2	foraging	* <i>Tursiops truncatus</i> (echolocation click train) * <i>Rousettus aegyptiacus</i> (echolocation click)
I	2	2	distress	* <i>Rousettus aegyptiacus</i> (isolation call) * <i>Phyllostomus hastatus</i> (infant isolation call)
I	2	2	distress	* <i>Rousettus aegyptiacus</i> (isolation call) * <i>Leptonychotes weddellii</i> (pup primary call)
I	2	2	other	* <i>Rousettus aegyptiacus</i> (echolocation click) * <i>Phyllostomus hastatus</i> (echolocation call)
I	2	2	cohesion	* <i>Saccopteryx bilineata</i> (maternal directive call) * <i>Phyllostomus hastatus</i> (infant isolation call)
I	2	2	cohesion	* <i>Saccopteryx bilineata</i> (maternal directive call) * <i>Leptonychotes weddellii</i> (pup contact call)
I	2	2	foraging	* <i>Saccopteryx bilineata</i> (echolocation call) * <i>Phyllostomus hastatus</i> (echolocation call)
I	2	2	danger	🦉 <i>Parus minor</i> (jar call) 🦉 <i>Gallus gallus</i> (ground-predator cackling alarm)
I	2	2	danger	* <i>Cercopithecus nictitans</i> (chirp) * <i>Marmota flaviventris</i> (whistle)

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Classes	Families	Species	Category	Members
I	2	2	cohesion	** <i>Leptonychotes weddellii</i> (pup primary call) ** <i>Marmota flaviventris</i> (pup scream)
I	I	2	cohesion	** <i>Loxodonta africana</i> (rumble) ** <i>Elephas maximus</i> (rumble)
I	I	2	foraging	** <i>Pan paniscus</i> (grunt) ** <i>Pan troglodytes</i> (grunt)
I	I	2	danger	** <i>Cercopithecus campbelli</i> (krak) ** <i>Chlorocebus pygerythrus</i> (chirp)
I	I	2	danger	** <i>Papio anubis</i> (cough-bark) ** <i>Macaca mulatta</i> (bark)
I	I	2	danger	** <i>Macaca mulatta</i> (bark) ** <i>Chlorocebus pygerythrus</i> (bark)

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H Reserve material

Figure 12, Figure 13, Figure 14, Figure 15.

[figure file acoustic_umap.png missing]

Figure 12. Acoustic embedding space (UMAP). Each point is a call in AnimalLex, positioned by acoustic description embedding and colored by taxonomic class. Clusters correspond to spectro-temporal call types (tonal long calls, broadband alarm bursts, etc.) that recur across classes, illustrating the convergent acoustic solutions captured by the database.

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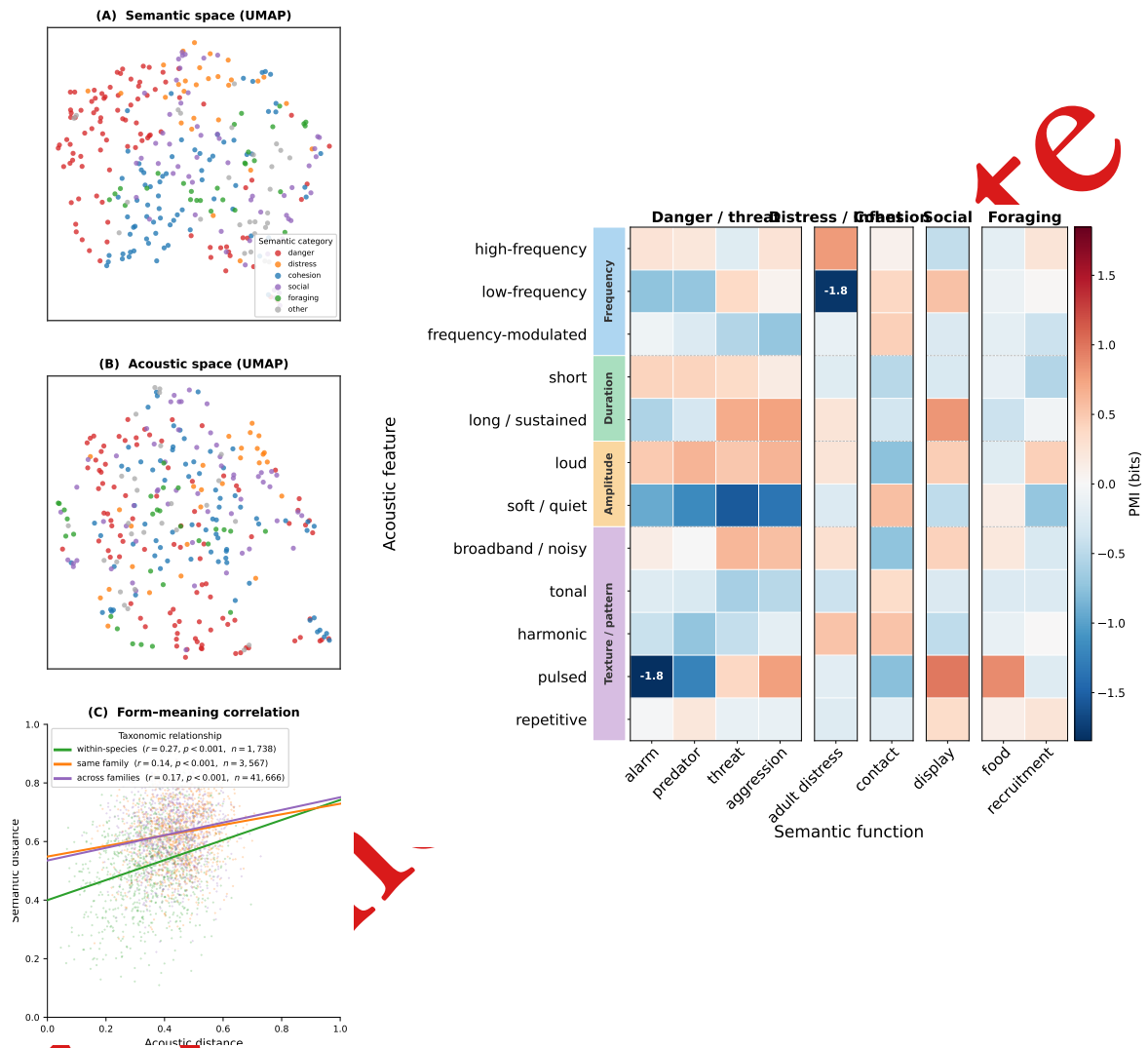


Figure 13. **[Draft] Merged Figs. 2+3 candidate.** Left column: **(A)** semantic UMAP, **(B)** acoustic UMAP, **(C)** Mantel scatter (same as Fig. 2). Right column: **(D)** PMI heatmap with semantic functions as rows and acoustic features as columns (square cells); rows and columns ordered by hierarchical clustering (same data as Fig. 4).

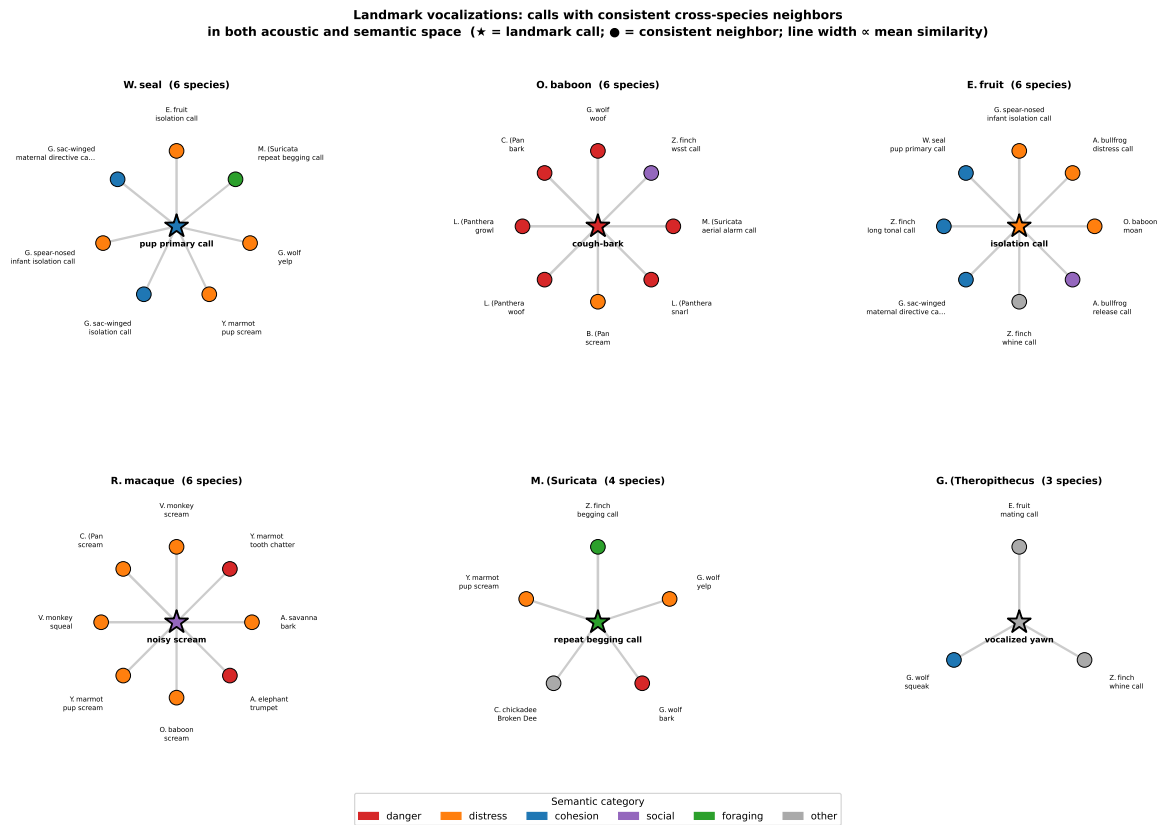


Figure 14. [Alternative to Fig. 15] Cross-species lexical translation. Calls with consistent cross-species neighbors in both acoustic and semantic space. Each rosette shows one landmark call (★, labeled with its call name at center) connected to up to eight calls from other species whose acoustic *and* semantic embeddings both rank among its top-20 nearest neighbors. Node color encodes semantic category (same scheme as Fig. 2); spoke width is proportional to mean acoustic–semantic similarity. Parenthetical counts give the number of distinct species with at least one consistent neighbor (range: 10–16). These landmark calls represent acoustic–semantic motifs independently co-opted across the tree of life—the concrete instances of the biological code quantified by the Mantel test (Fig. 2C).

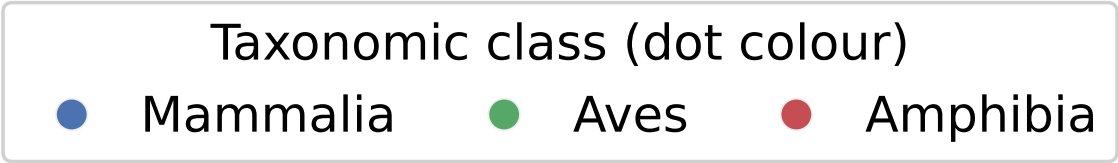


Figure 15. Cross-species call groups sharing acoustic and semantic form. Each panel shows one group of calls—one per species—belonging to a single semantic category (dark header). A pair of calls qualifies iff each is simultaneously the top-1 acoustic *and* top-1 semantic nearest neighbour of the other within its counterpart species; groups are cliques in this species-pairwise mutual-consistency graph, selected to maximise taxonomic diversity (class and family spread; see Supplementary Sections C and G). The coloured dot encodes taxonomic class (legend). *Top:* 2×2 layout; each row shows species and call name, followed by semantic (*Sem:*) and acoustic (*Acc:*) descriptions. *Bottom:* same four groups arranged horizontally; each row shows common name (line 1), call name (line 2), semantic keywords (line 3), and acoustic description in italics (line 4).